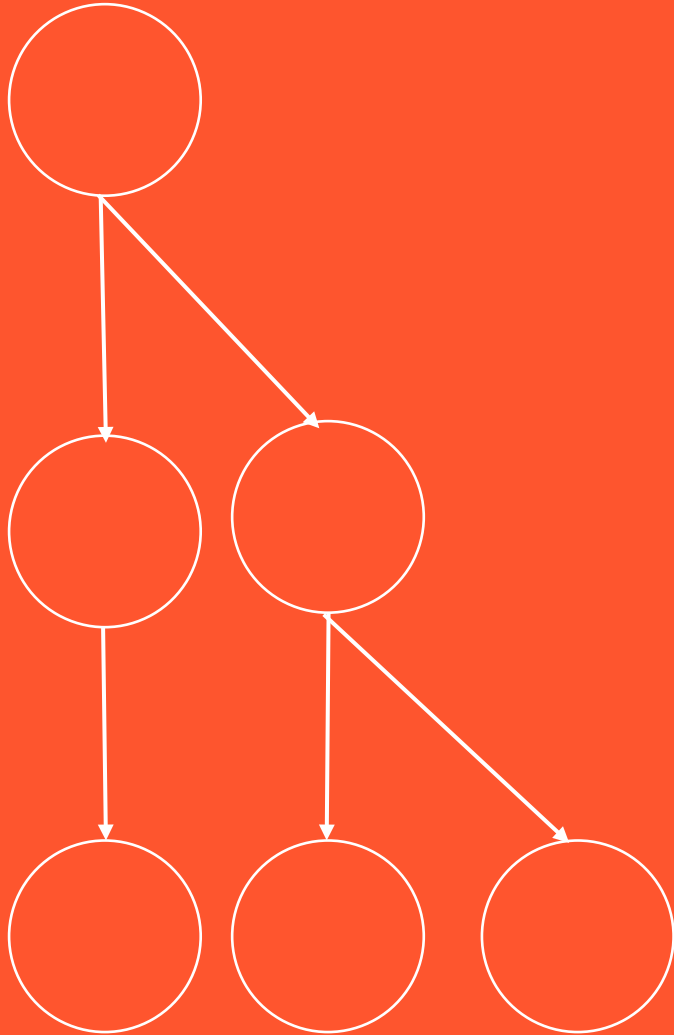


BCOG 100

The Evolution of Neurons

Lecture 2: The Evolution of Neurons



UNIVERSITY OF
ILLINOIS
URBANA-CHAMPAIGN

Reading Check 4-2

Password: action

Question: What does "all or none" mean with regard to neurons, and why is it useful for neurons to be that way?



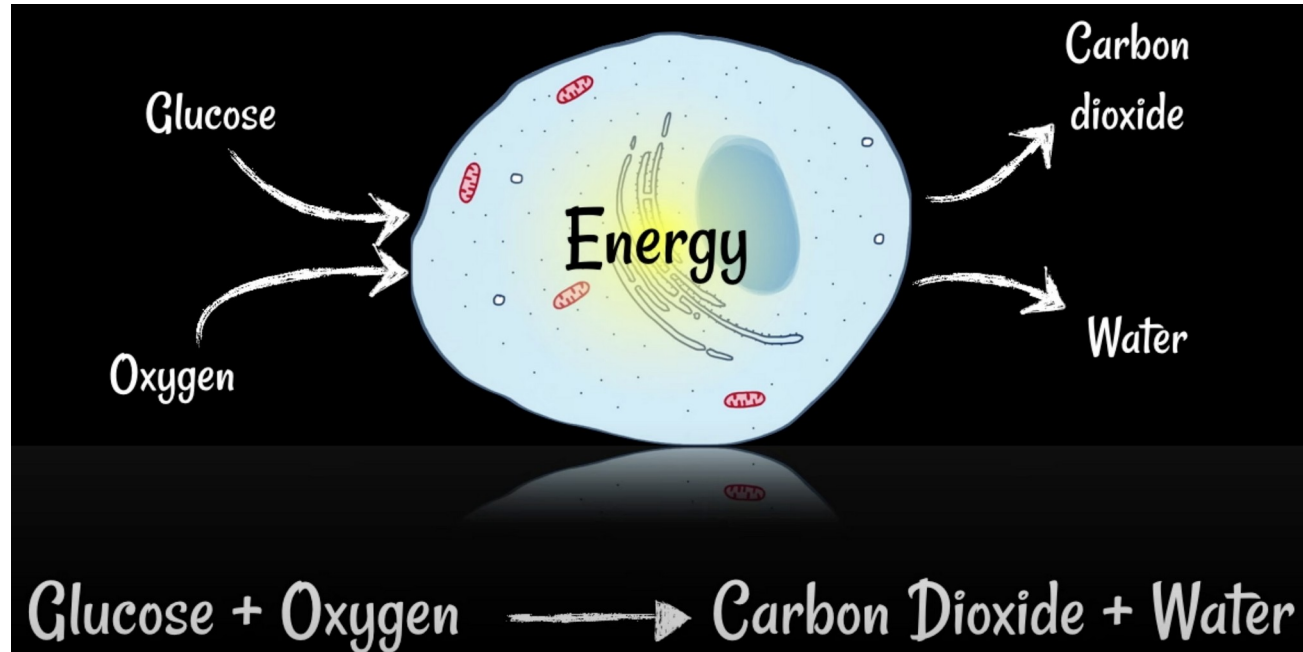
The Evolution of Neurons

Today's Outline

- Oxygen → energy revolution
- Eukaryotes & endosymbiosis
- Multicellularity & specialization
- Animals vs. plants vs. fungi
- Why animals needed neurons
- What neurons are & how they work



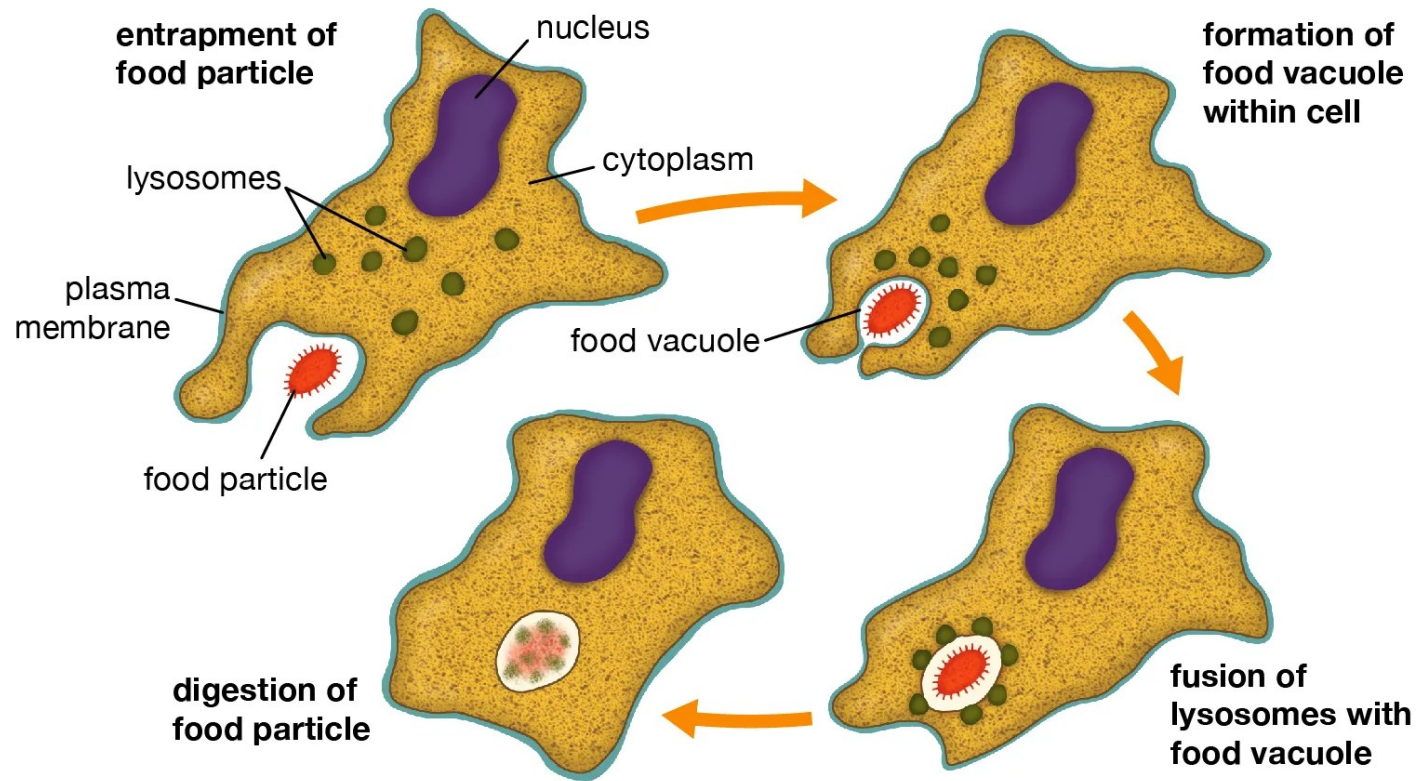
The Evolution of Neurons



After Oxygen: A New Energy World

- Respiration with O_2 = $\sim 15\times$ more ATP
- Energy surplus $\rightarrow$ new possibilities:
 - bigger cells
 - faster growth and replication
 - more complex internal machinery

The Evolution of Neurons



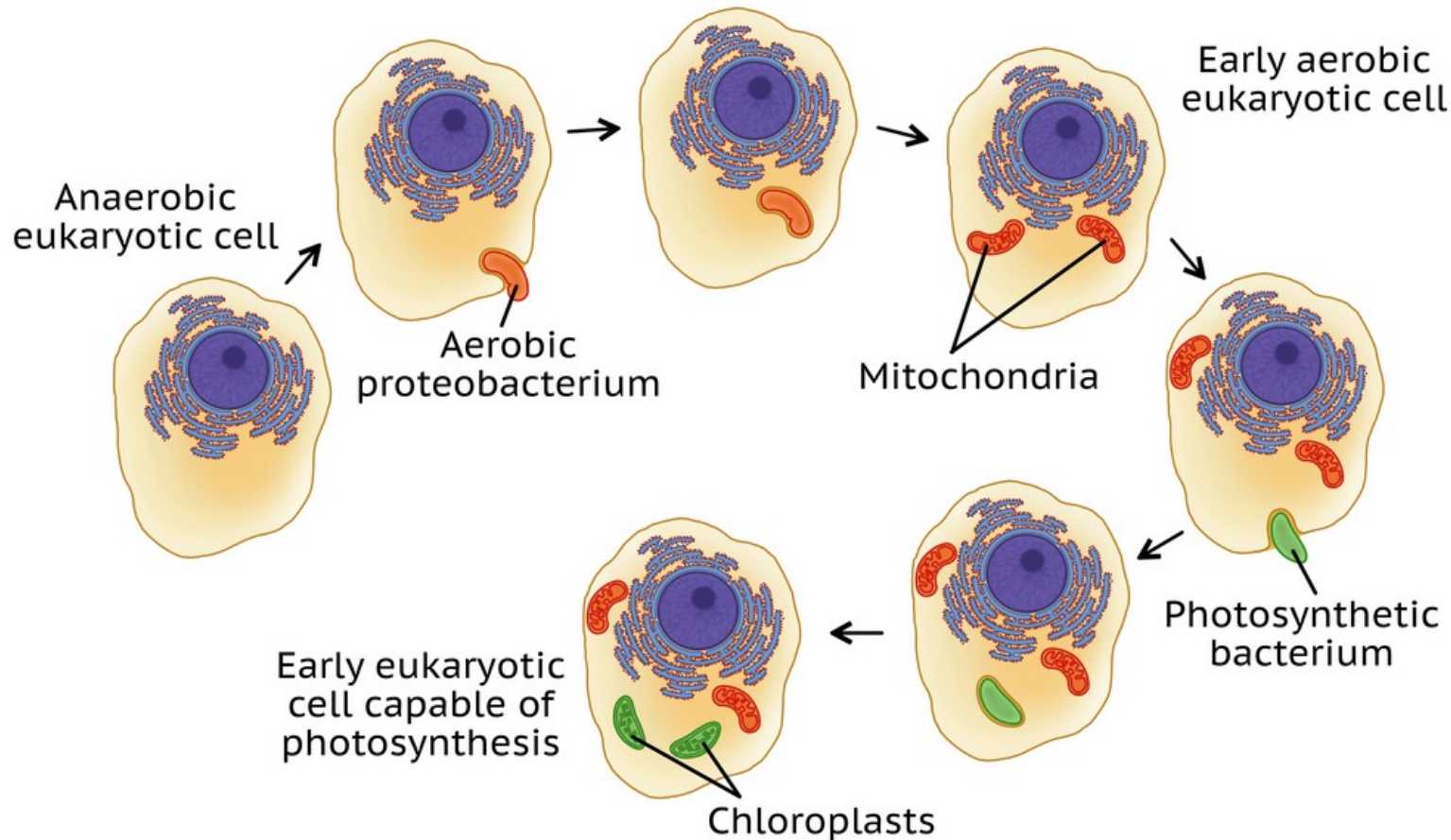
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Predator-Prey Arms Race

- More energy → new lifestyles
- Hunting emerges
- Predators vs. prey = speed & coordination matter

The Evolution of Neurons

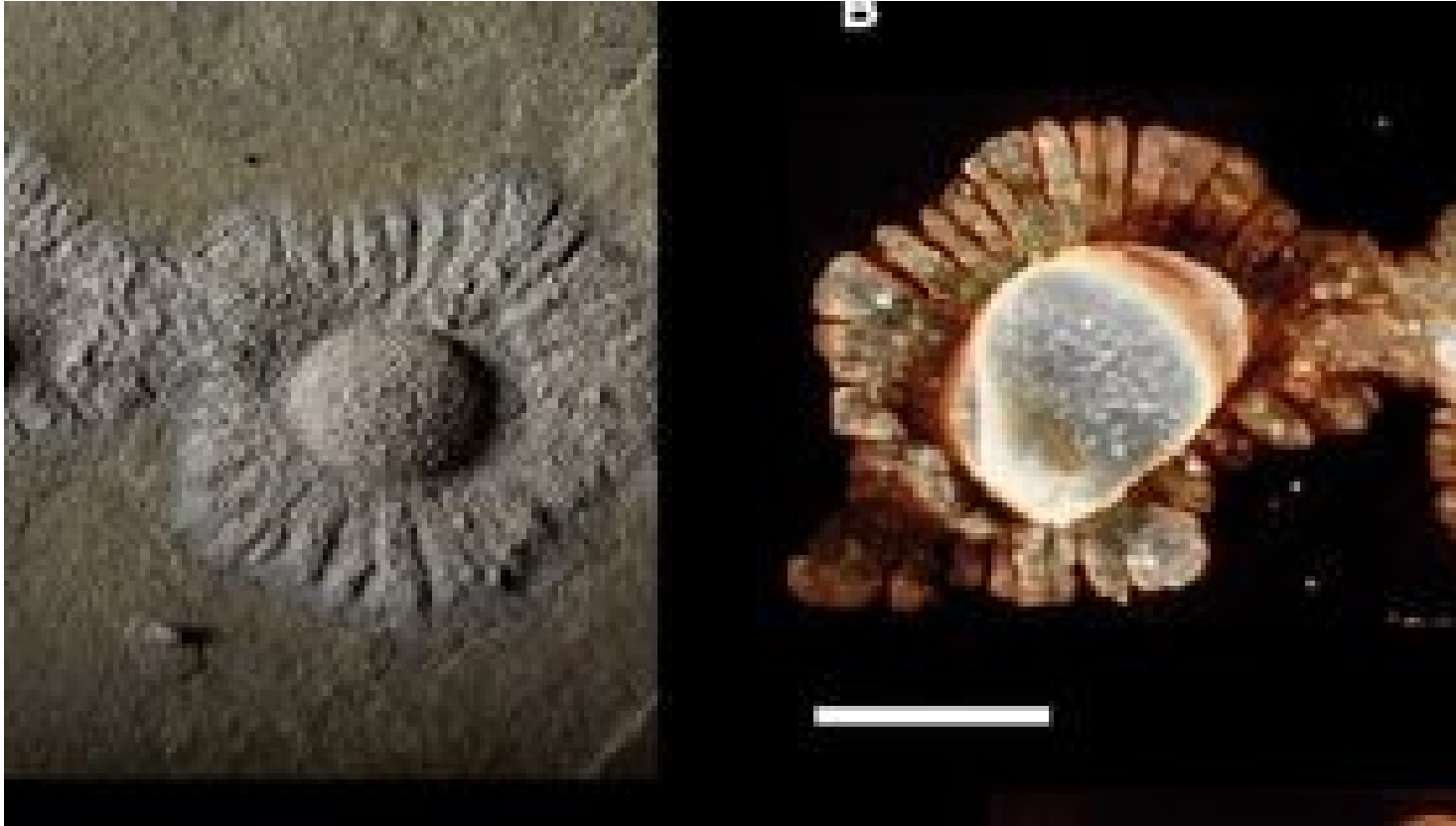
THE ORIGIN OF MITOCHONDRIA AND CHLOROPLASTS



Eukaryotes: Bigger, More Complex Cells

- Endosymbiosis = “cells inside cells”
- Engulfed bacteria → mitochondria
- Advantages: more energy, more complexity

The Evolution of Neurons



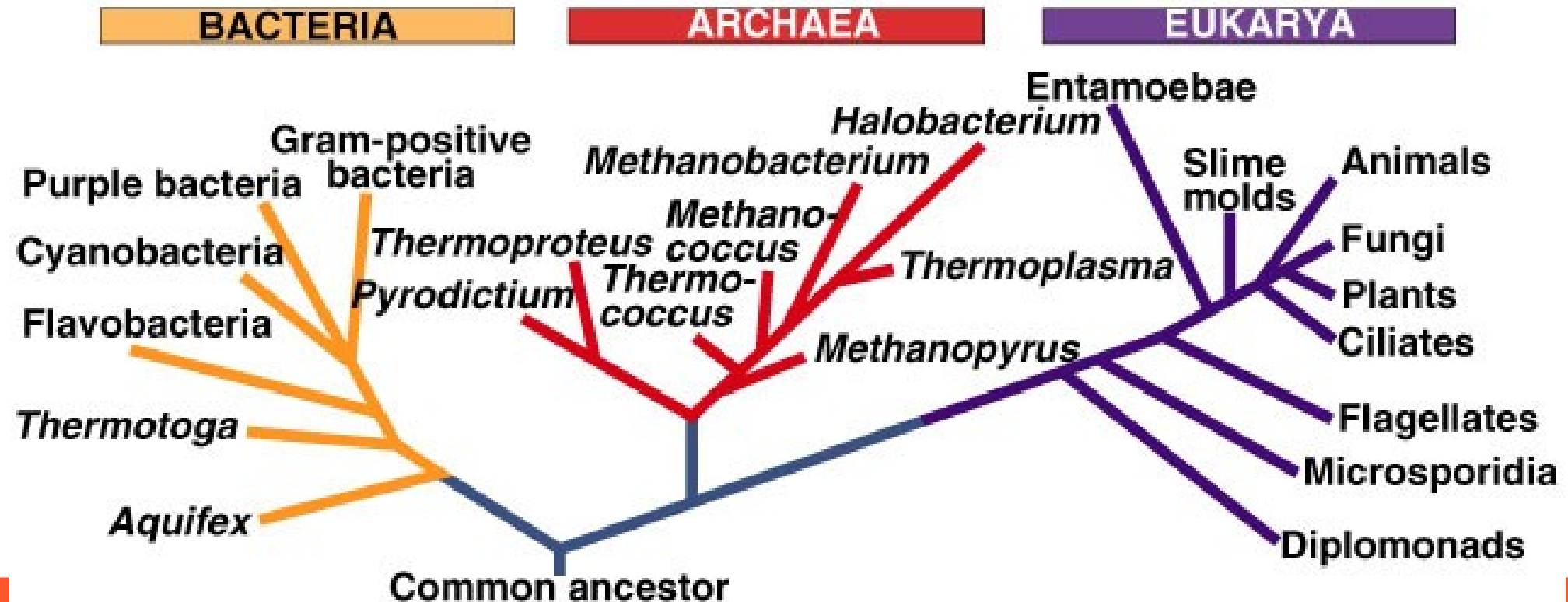
Multicellularity: Size and Specialization

- Bigger is better (avoid predators, catch prey)
- Specialization = division of labor
- Cells → tissues → organs

The Evolution of Neurons

Three Strategies of Complex Life

- Plants: photosynthesis + chloroplasts
- Fungi: decompose, absorb (no neurons)
- Animals: active hunting (need neurons)



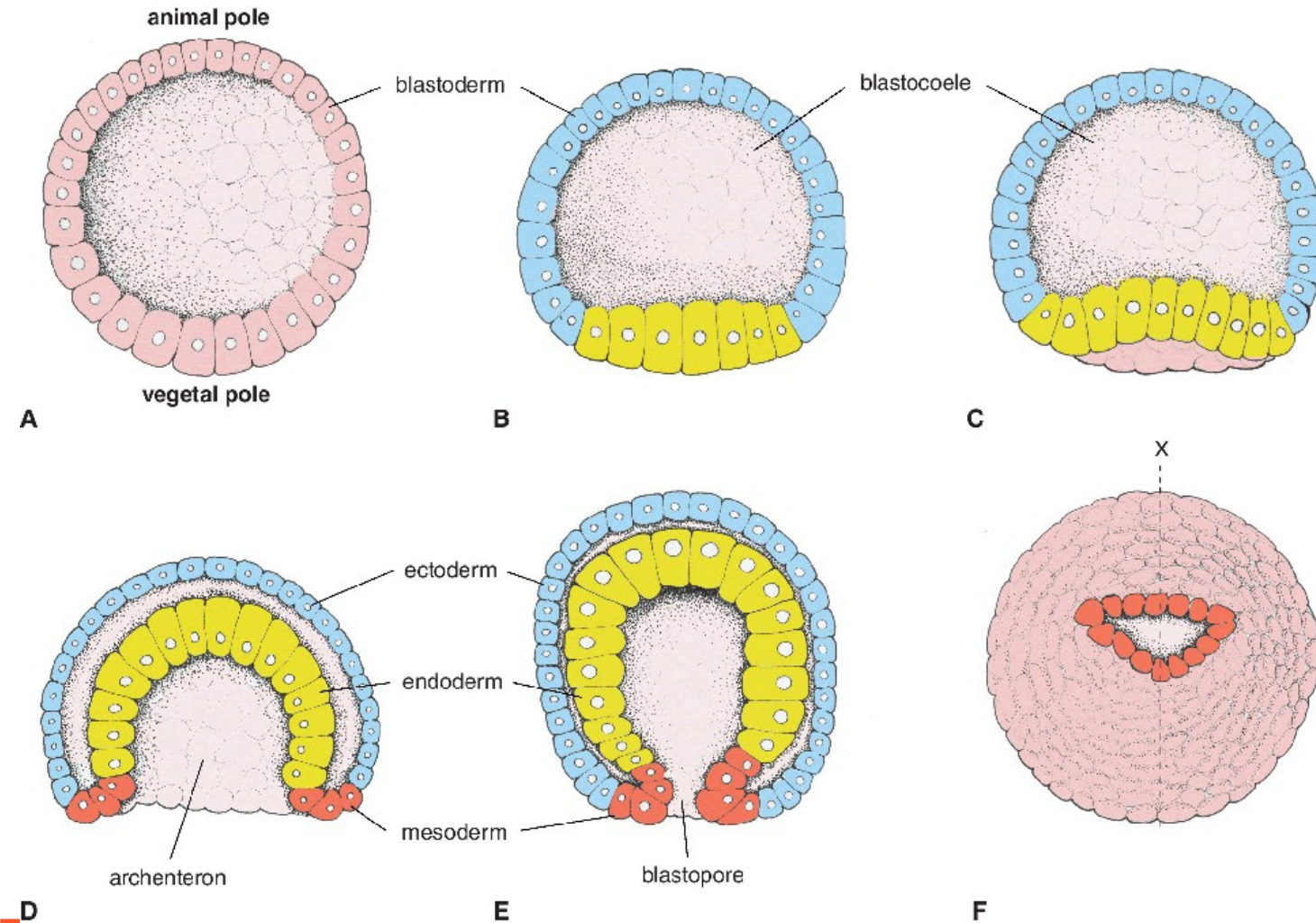
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Why Animals Benefited from Neurons

- Plants: no need to chase
- Fungi: no need to move
- Animals: speed = survival
- Sense → Decide → Act

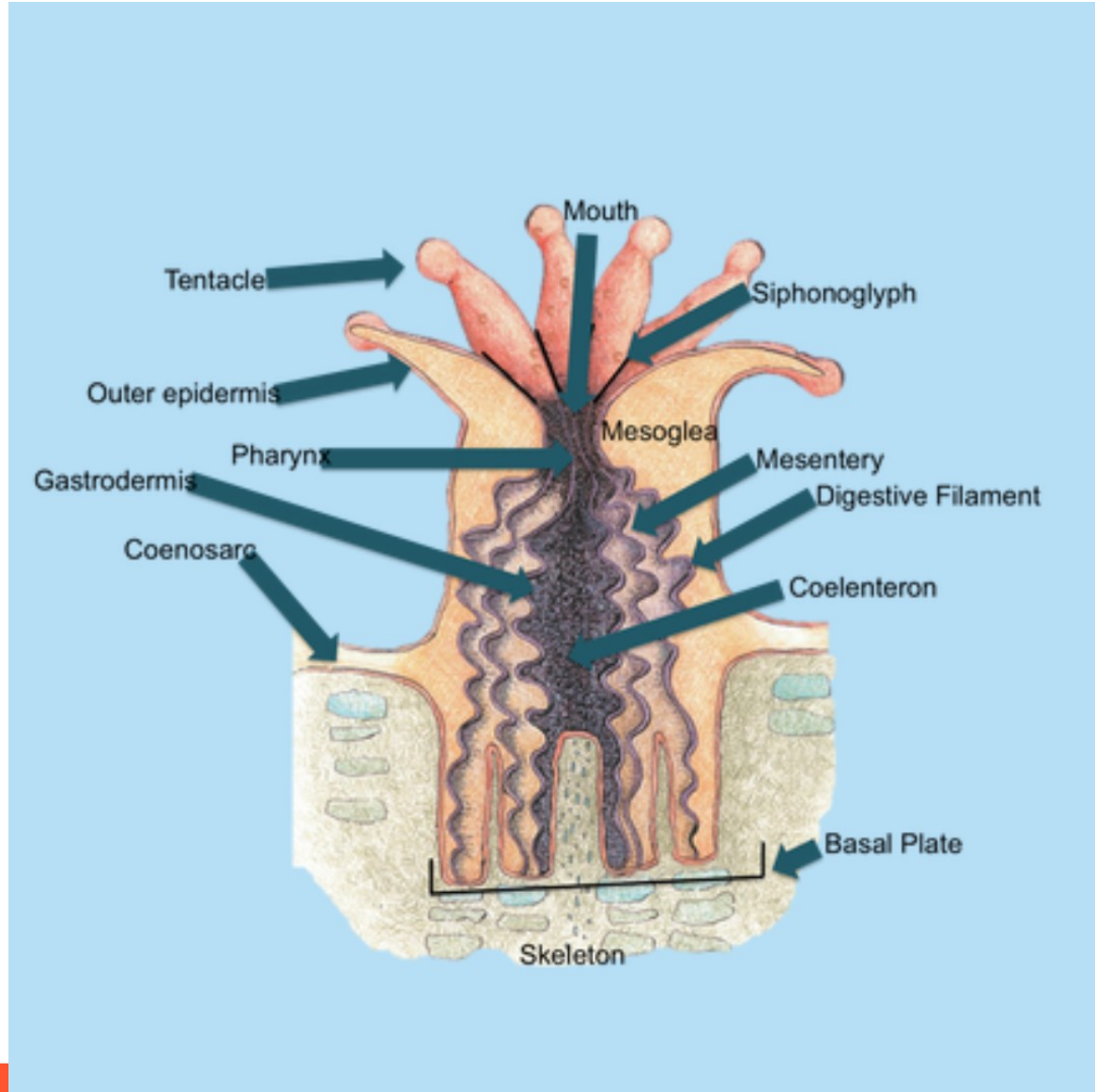
The Evolution of Neurons



Gastrulation & the Animal Template

- Gastrulation: early animal development creating a cavity for digestion
- Three key tissues emerge:
 - Gut (stomach)
 - Muscles
 - Neurons

The Evolution of Neurons



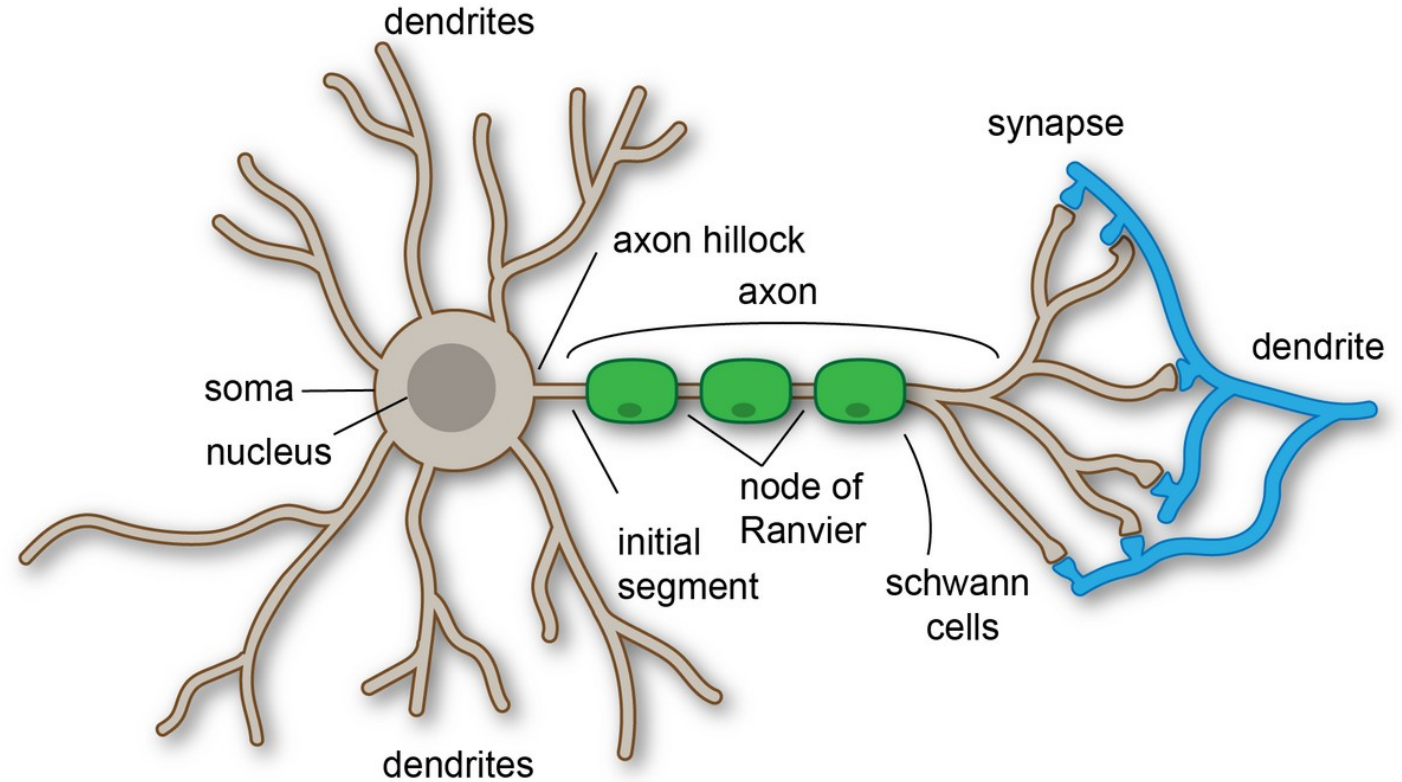
Earliest Nervous Systems

- Earliest animals: corals, jellyfish
- **Nerve nets** = simple webs of neurons
- No brain, but coordinate muscles & feeding

Coral Eating and Digestion

- Tentacles sense prey (touch, chemicals)
- Neurons trigger stinging cells (cnidocytes)
- Muscles contract → tentacle pulls prey in
- Gut digests food

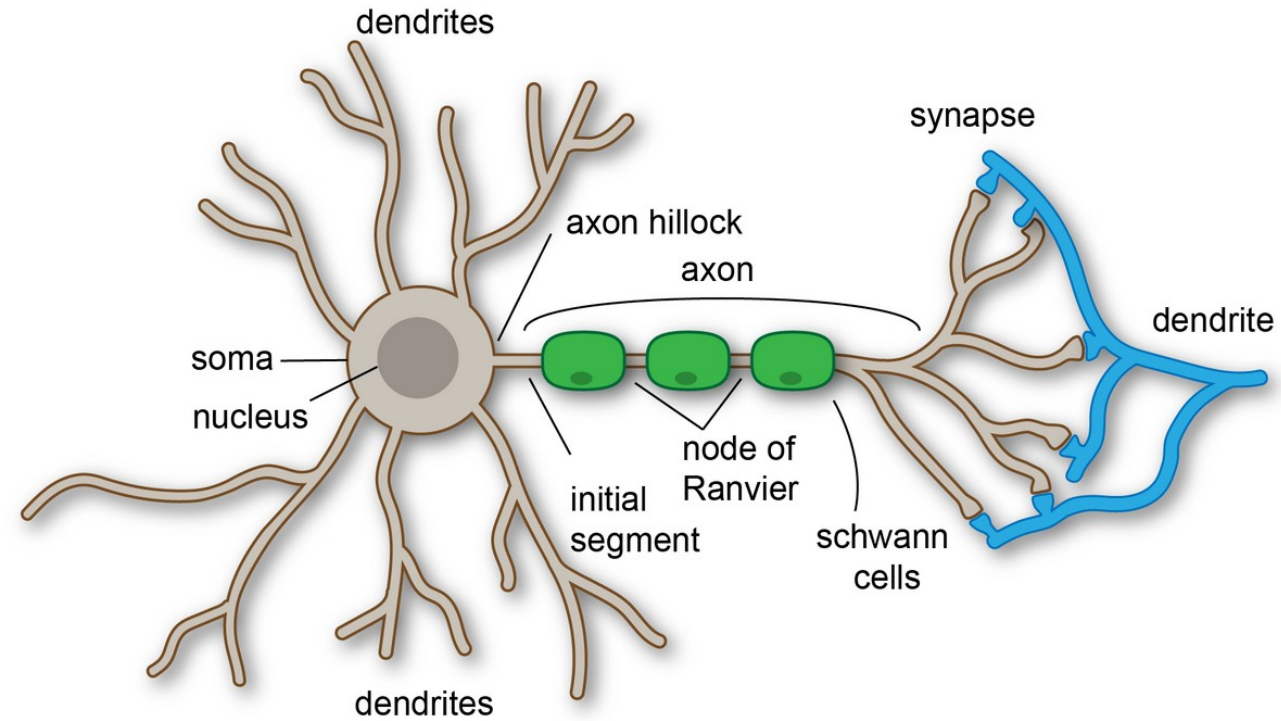
Neuron Structure and Function



What is a Neuron?

- Cells already communicated (e.g., chemicals, hormones)
- Neurons = faster, longer-distance, more precise
- Receive → Process → Send signals

Neuron Structure and Function



The Parts of a Neuron

- **Dendrites:** receive signals
- **Soma (cell body):** integrates
- **Axon:** sends signal
- **Synapse:** junction to next cell

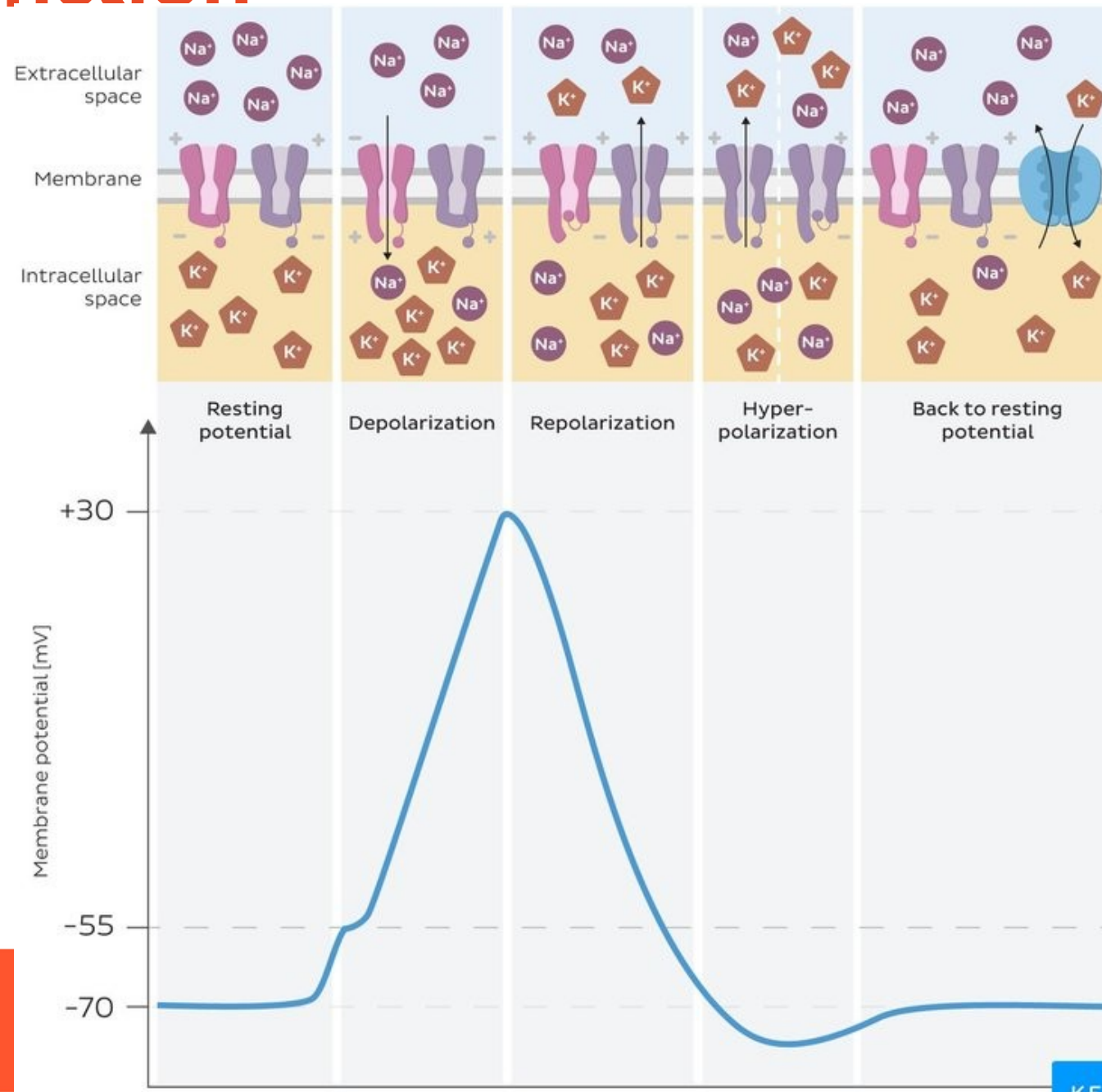
Neuron Structure and Function

Action Potentials

- Electrical impulses = **spikes**
- All-or-nothing
- Travel long distances quickly

Why Spikes Matter

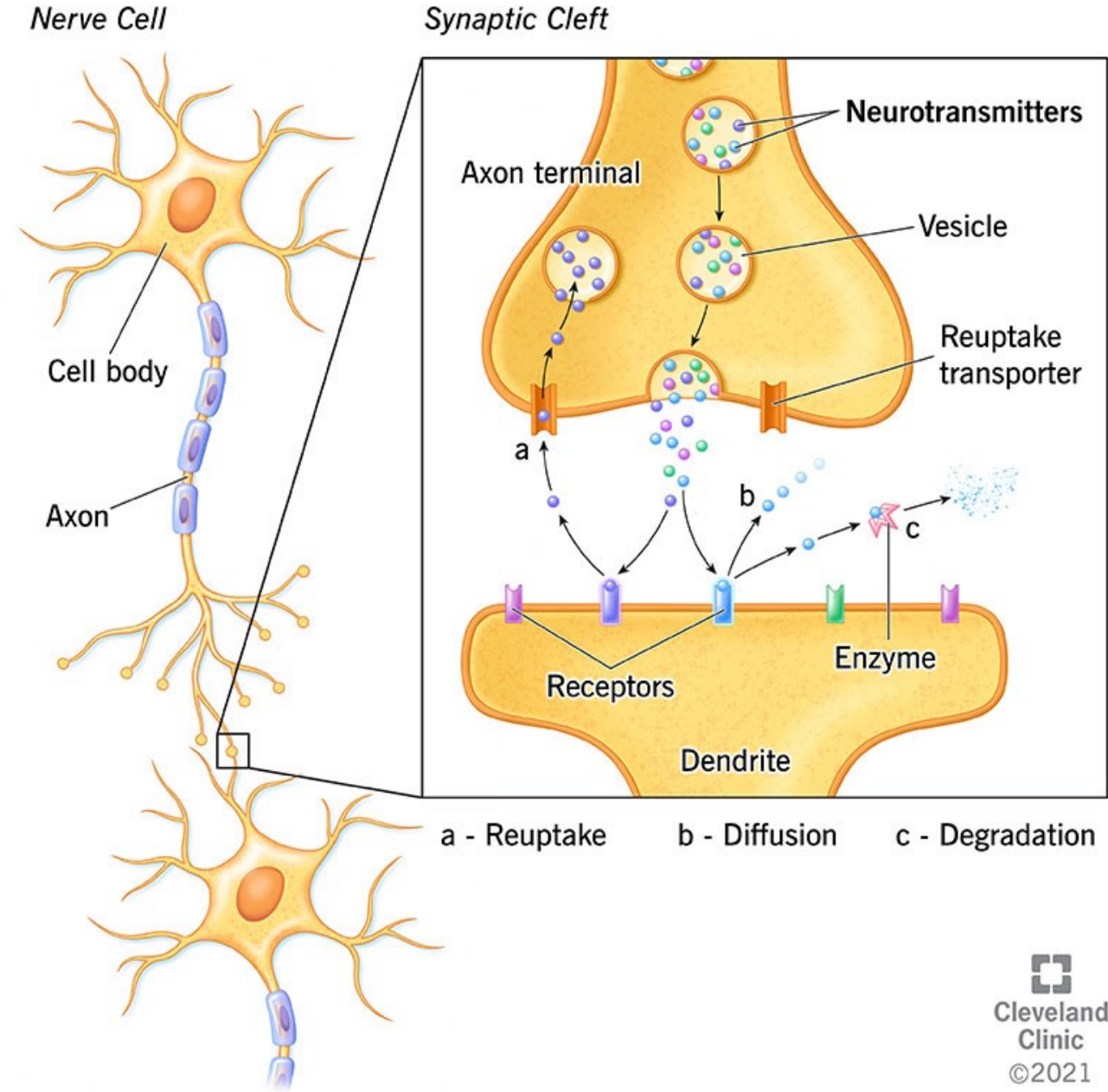
- Fast → millisecond timing
- Reliable → signals don't fade
- Coordinate whole body
- Foundation for circuits & brains



Neuron Structure and Function

Neurotransmitters: Signals Between Neurons

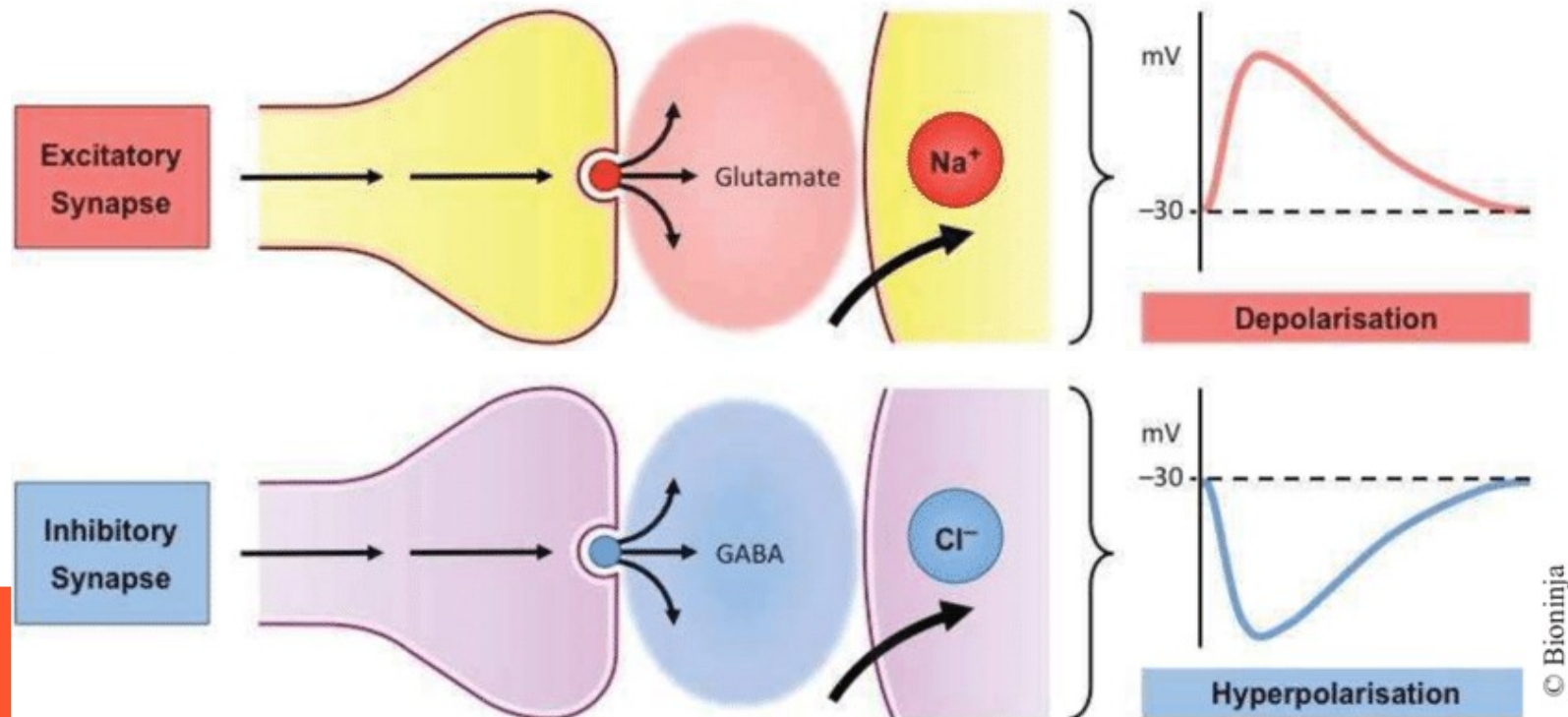
- Spikes → release chemicals at **synapse**
- Neurotransmitters cross gap
- Bind to receptors on next neuron
- Different transmitters = different effects
 - **Glutamate**: excitatory.
 - **GABA**: inhibitory
 - **Dopamine, serotonin, acetylcholine, norepinephrine** — these have more specialized roles, influencing things like motivation, mood, memory, or alertness



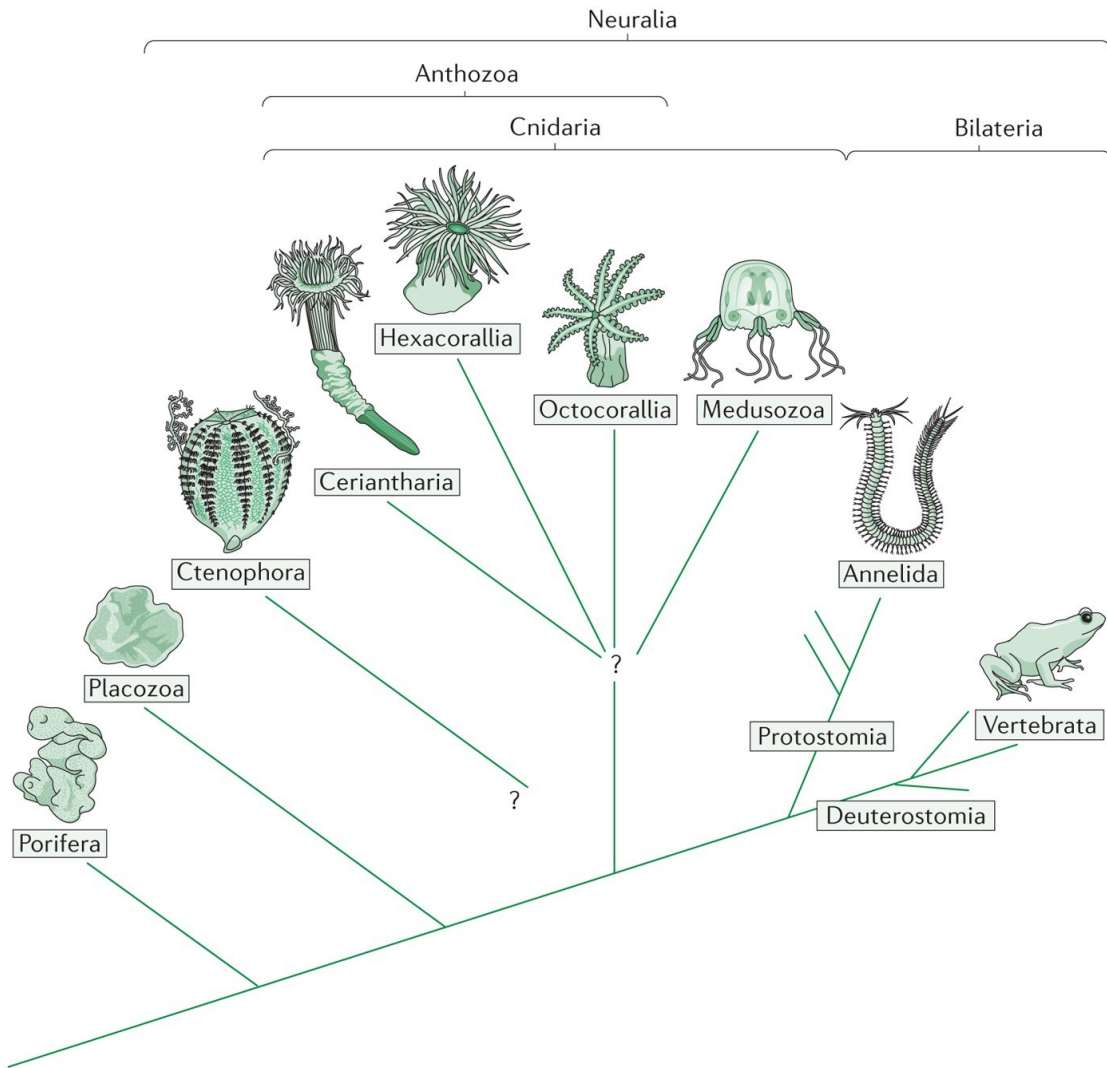
Neuron Structure and Function

Excitatory vs. Inhibitory Neurons

- **Excitatory:** increase chance of downstream (efferent) neurons firing (main example Glutamate).
- **Inhibitory:** decrease chance of downstream (efferent) neurons firing (main example GABA)
- **Neural Circuits:** mixtures of excitatory and inhibitory neurons that allow for specific computations to occur.



Summarizing the Evolution of Neurons



Nature Reviews | Neuroscience

Today we saw...

- Oxygen → new energy → eukaryotes
- Multicellularity → specialization
- Animals needed speed → neurons evolved
- Neurons = spikes, synapses, neurotransmitters

Next time...

- Nerve nets are not yet brains
- How neurons were organized to steer, choose, decide
- The first simple brains